PROBABILITY & STOCHASTIC PROCESSES

I) BASICS [SET THEORY]

Let sample space = S
experiment = E
event = A
} then P(A) ⇒ probability of A

- $0 \leq P(A) \leq 1$
- $P(S) = 1$

① Complement

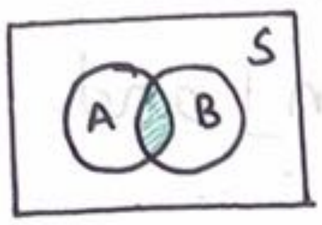
$A' = S \setminus A$ =

(all elements of S that are not in A)



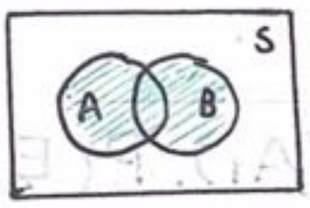
② Intersection

$A \cap B$ =



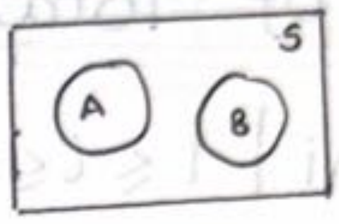
③ Union

$A \cup B$ =

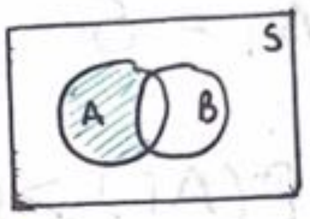


④ Mutually exclusive

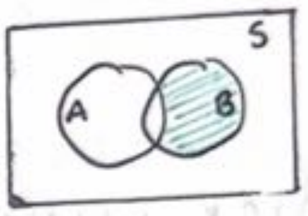
$A \cap B = \phi$ (empty set)



⑤ $A \setminus B$ (A complement B)

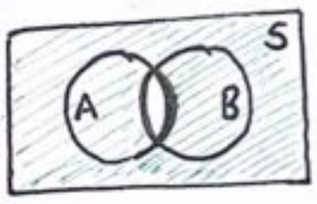


$B \setminus A$ (B complement A)

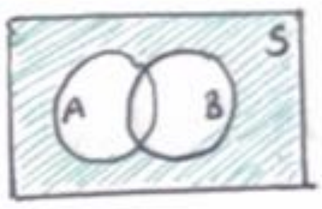


De Morgan Laws

⑥ $(A \cap B)' = A' \cup B'$



$(A \cup B)' = A' \cap B'$



(mutually exclusive) $S = S \cup \phi$ (S: everything, ϕ: nothing)

$P(S) = P(S) + P(\phi)$

$\therefore \boxed{P(\phi) = 0}$

if $A_1, A_2 \ldots A_n$ are (pairwise) mutually exclusive, then $P(A_1 \cup A_2 \cup A_3 \ldots A_n)$

$$= \sum_{i=1}^{n} P(A_i) = P(A_1) + P(A_2) + \ldots P(A_n)$$

Independent Events ⇒ $P(A \cap B) = P(A) \cdot P(B)$

⑥ $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

⑦ $P(A \cup B \cup C)$
$= P(A) + P(B) + P(C)$
$- P(A \cap B) - P(B \cap C) - P(C \cap A)$
$+ P(A \cap B \cap C)$

Properties:

① $P(\phi) = 0$

② if $A \subseteq B$, $P(A) \leq P(B)$

$B = A \cup (A' \cap B)$

$P(B) = P(A) + P(A' \cap B)$, $P(A' \cap B) \geq 0$

$\therefore P(A) \leq P(B)$

③ if $A \cap B = \phi$, then $P(A) \leq P(B')$

$A \subseteq B'$

$P(A) < P(B')$ by ②

④ if A is an event in S then $P(A') = 1 - P(A)$

$A' \cup A = S$

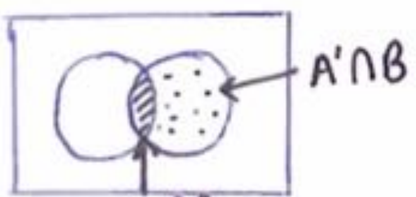
$P(A') + P(A) = P(S)$

$P(A') = 1 - P(A)$

⑤ $P(A' \cap B) = P(B) - P(A \cap B)$

$B = (A' \cap B) \cup (A \cap B)$

$P(A' \cap B) = P(B) - P(A \cap B)$



⑧ if $B \subseteq A'$

$P(A \cap B') = P(A) - P(B)$



⑨ If A & B are independent
Then A & B' are also independent

⑩ If A & B are independent
A', B' are also independent

- **Conditional Probability**

$$P(B|A) = \frac{P(B \cap A)}{P(A)}$$

(probability of B given A)

or $P(B|A) \cdot P(A) = P(B \cap A)$

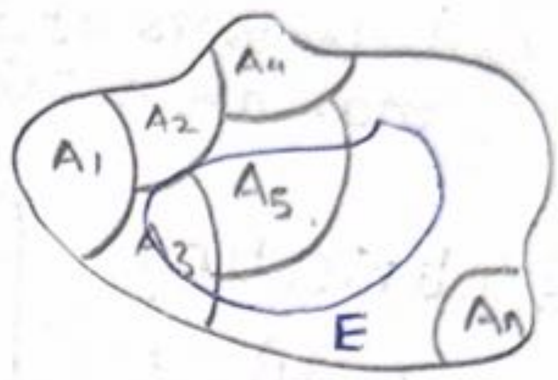
; $P(A) \neq 0$

Theorems:

1) Law of Total Probability

Let $\{A_i \mid 1 \leq i \leq n\}$ be a partition of a sample space S corresponding to a random experiment i.e $A_i \cap A_j = \phi$ $[1 \leq i \neq j \leq n]$ and

$$\bigcup_{i=1}^{n} A_i = S$$



Further suppose, $P(A_i) > 0$ $(1 \leq i \leq n)$

Then for any event E in S,

$$P(E) = P(A_1) \cdot P(E|A_1) + P(A_2) \cdot P(E|A_2) + \vdots + P(A_n) \cdot P(E|A_n)$$

Proof:

① we know $A_i \cap A_j = \phi$ $[1 \leq i \neq j \leq n]$ [$\because$ A_i is a partition and A_i, A_j are mutually exclusive (pairwise)]

② $\therefore A_1 \cup A_2 \cup A_3 \ldots \cup A_n = S$

$P(A_1) + P(A_2) + \ldots P(A_n) = P(S)$

③ for event E, $E = E \cap S$

$\therefore E = E \cap (A_1 \cup A_2 \cup \ldots A_n)$ (from ②)

$\therefore E = (E \cap A_1) \cup (E \cap A_2) \ldots \cup (E \cap A_n)$ [De Morgan Laws]

④ $\therefore P(E) = P(E \cap A_1) + P(E \cap A_2) + \ldots P(E \cap A_n)$

$P(E) = P(E|A_1) \cdot P(A_1) + P(E|A_2) \cdot P(A_2) + \ldots P(E|A_n) \cdot P(A_n)$ [from cond. prob formula]

Hence proved

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② Baye's Theorem

Let $\{E_1, E_2, E_3 \ldots, E_k\}$ be a partition of sample space S corresponding to the random experiment.
Then for event F in S with $P(F) > 0$:

$$P(E_j|F) = \frac{P(F|E_j) \cdot P(E_j)}{P(E_1) \cdot P(F|E_1) + P(E_2) \cdot P(F|E_2) + \ldots P(E_k) \cdot P(F|E_k)}$$

I) 1-D Random variables

(imp) ⇒ Mean of a random variable might NOT EXIST i.e $\mu = 0$.

Discrete

① numerically valued function on discrete sample space S, corresponding to a random exp. E

where prob. is concentrated.

② Probability Distribution Function

$P\{X = X_i\} = P_i, \; i \geq 1$

$\sum_{i=1}^{n} P(X = X_i) = 1$

③ Expectation / Mean
(expected value of d·r·v X)

$$\mu / E(x) = \sum_{i=1}^{\infty} x_i P(x_i)$$

whenever this ∞ series is convergent

④ Variance
(spread from its mean)

$$E(x-\mu)^2 = \sum_{i=1}^{\infty} (x_i - \mu)^2 P(x_i) = V(x)$$

$$V(x) = E(x^2) - [E(x)]^2$$

⑤ Cumulative Distribution Function
[total probability accumulated upto a point]
how much accumulated

Discrete: $F(a) = \sum_{\forall x \leq a} P(x)$

Continuous: $F(a) = \int_{-\infty}^{x=a} f(x)\,dx$

Continuous

① a random variable which takes non-countable infinite number of values is called a c·r·v

② Probability Distribution Function

X = c·r·v
$f(x) \geq 0$
$\int_{-\infty}^{\infty} f(x)\,dx = 1$

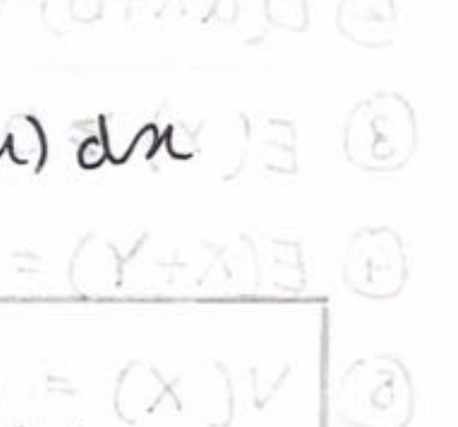
$$P(a \leq x \leq b) = \int_a^b f(x)\,dx$$

If r: real number
then, $P(X \geq r) = \int_r^{\infty} f(x)\,dx$

$$P(X \leq r) = 1 - \int_r^{\infty} f(x)\,dx$$

③ Expectation / Mean

$$E(x) = \int_{-\infty}^{\infty} x f(x)\,dx$$



④ Variance

$$V(x) = E(x^2) - [E(x)]^2$$

$$= \int_{-\infty}^{\infty} x^2 f(x)\,dx - \left[\int_{-\infty}^{\infty} x f(x)\,dx\right]^2$$

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III) Bernoulli Trials

Two outcomes → success
Two outcomes → failure

If a Bernoulli's Trial is performed n times with r: success and probability p & q respectively
n-r: failure

no. of trials → n, no. of success → r

$$^{n}C_r \, p^r \, q^{n-r} \quad , q = 1-p$$

p: prob of success; q: prob of failure; $n-r$: no. of failure

Note:

① for T/F ques
$P(\text{correct}) = \frac{1}{2}$
$P(\text{wrong}) = \frac{1}{2}$

② for n choices one correct
$P(\text{correct}) = \frac{1}{n}$

③ for n choices multiple correct
$P(\text{correct}) = \frac{1}{2^n - 1}$

Note:
if game is fair ?
means:
mean / avg. gain i.e $E(x) = 0$

Note: more basic properties of mean $[E(x)]$

① $E(X) = c$ if $P(X=c) = 1$, for any constant c

② $E(aX+b) = aE(X) + b$, for a, b constants

③ $E(X) \geq 0$, if X non-negative

④ $E(X+Y) = E(X) + E(Y)$

⑤ $V(X) = E(X^2) - [E(X)]^2$

mean of a random variable might NOT EXIST (i.e $\bar{x} = 0$).

V Two Dimensional Random Variable.

Let S be a sample space associated with random experiment E.

Let $X = X(s)$ be two fractions each
$Y = Y(s)$ assigning a real number to each outcome $s \in S$

Then the pair $(X, Y) \Rightarrow$ 2 dim r.v
eg: height, weight.

#Note:
if $X_1 = X_1(s), X_2 = X_2(s) \ldots X_n = X_n(s)$
are n-functions assigning a real number to every outcome $s \in S$, then
$X_1, X_2, \ldots X_n$ are n-dim r.v

DISCRETE

① X, Y is discrete if:
$X = x_1, \ldots$ } finite or
$Y = y_1, \ldots$ } countably ∞

② Prob. distribution function

$(x_i, y_j, P(x_i, y_j))_{i=1,j=1}^{\infty}$ dis

X	. . .
Y	. . .
f(x,y)	. . .

$P(x_i, y_j) = P(X = x_i, Y = y_j)$ should satisfy

① $P(x_i, y_i) \geq 0$

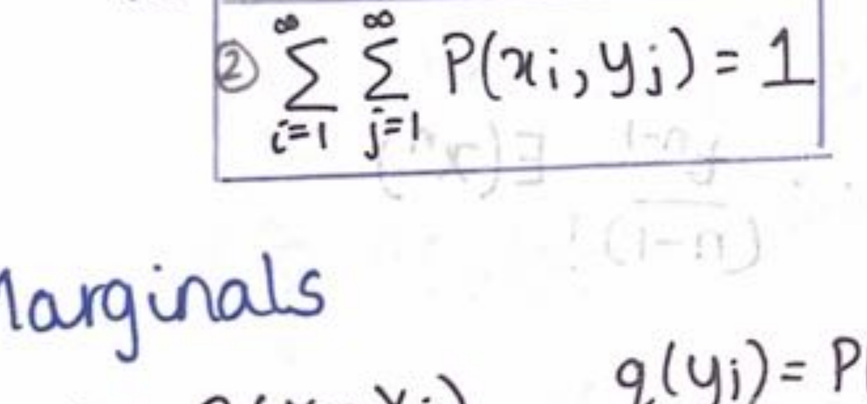
② $\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} P(x_i, y_j) = 1$

③ Marginals

$P(x_i) = P(X = x_i)$ $\quad q(y_i) = P(Y = y_j)$
$= P(X = x_i, Y = y_1, y_2 \ldots, y_n)$

$P(x_i) = \sum_{j=1}^{\infty} P(x_i, y_j)$ $\quad y \to$ variable, $x \to$ fixed

$q(y_i) = \sum_{i=1}^{\infty} P(x_i, y_j)$ $\quad x \to$ variable, $y \to$ fixed.

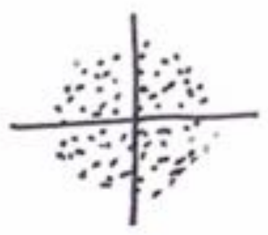


④ Conditional Prob.

$P(x_i | y_i) = P(X = x_i | Y = y_i)$
$= \frac{P(X = x_i, Y = y_i)}{P(Y = y_i)}$

CONTINUOUS

① X, Y is continuous if:
X, Y assumes all values in Euclidean Plane (uncountably ∞)



② Prob. Density Function $f(x, y)$ should satisfy:

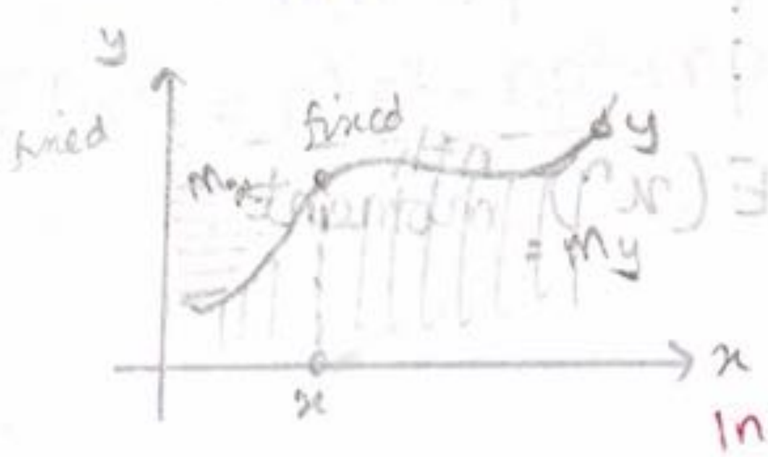
① $f(x, y) \geq 0 \quad \forall \; x, y \in \mathbb{R}$

② $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)\, dx\, dy = 1$

③ Marginals

$g(x) = \int_{-\infty}^{\infty} f(x, y)\, dy$ (x fixed)

$h(y) = \int_{-\infty}^{\infty} f(x, y)\, dx$ (y fixed)



x, y independent
In case of independence, marginal ≡ conditional

④ Conditional Density

$g(x|y) = \frac{f(x,y)}{h(y)}$ $\quad h(y|x) = \frac{f(x,y)}{g(x)}$

$h(y) > 0$ $\quad g(x) > 0$

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Moment Generating Function

Let X : d·r·v with p·d·f $P(x_i) = P(X = x_i)$ $i = 1, 2, 3 \ldots$

mgf of x is defined as:

$$M_x(t) = \sum_{j=1}^{\infty} e^{tx} p(x_j) \quad , \quad \text{for } h > 0, \; -h < t < h$$

↑ expected value of e^{tx}

$$= E(e^{tx})$$

$$M_x(t) = E(e^{tx})$$

$$= E\left(1 + tx + \frac{t^2x^2}{2!} + \frac{t^3x^3}{3!} + \ldots\right)$$

$$= E(1) + E(tx) + E\left(\frac{t^2x^2}{2!}\right) + \ldots$$

$$= 1 + tE(x) + \frac{t^2}{2!}E(x^2) + \ldots + \frac{t^n}{n!}E(x^n)$$

$\therefore E(x^n) \Rightarrow$ coefficient of $\frac{t^n}{n!}$ in expansion of $M_x(t)$

$$M_x'(t) = 0 + E(x) + tE(x^2) + \frac{t^2}{2}E(x^3) + \ldots \frac{t^{n-1}}{(n-1)!}E(x^n)$$

$t = 0 \Rightarrow \boxed{E(x)}$ (1st moment)

$$M_x''(t) = 0 + 0 + E(x^2) + t(E(x^3)) + \ldots \frac{t^{n-2}}{(n-2)!}E(x^n)$$

$t = 0 \Rightarrow \boxed{E(x^2)}$ (2nd moment)

$\vdots$

$M_x^n(0) = E(x^n)$ n^{th} moment.

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I. Binomial Distribution (Biparametric)

Let X be a d·r·v taking values 0, 1, 2 . . .

$0 < p < 1$ (success)

$q = 1 - p$ (failure)

$P(x) = {}^{n}C_x\, p^x q^{n-x}$ probability distribution of x

p·d·f

① $P(X) \geq 0$

② $\sum_{i=1}^{n} P(x_i) = 1$ $\left[\sum_{x=1}^{n} {}^{n}C_x\, p^x q^{n-x} = (p+q)^n = 1^n = 1\right]$

mean = np

variance = npq

Proof: $n \rightarrow$ trials
$p \rightarrow$ prob. success
$q \rightarrow$ prob. failure
$x \rightarrow 0, 1, 2, 3 \ldots n$

$P(x) = {}^{n}C_x\, p^x q^{n-x}$

① $M_x(t) = \sum_{x=0}^{n} e^{tx} P(x)$

$= \sum_{0}^{n} e^{tx}\, {}^{n}C_x\, p^x q^{n-x}$

$= \sum_{0}^{n} {}^{n}C_x (pe^t)^x\, q^{(n-x)}$

$= (pe^t + q)^n$

② $M_x'(t) = n(pe^t + q)^{n-1} \cdot pe^t$

$M_x'(0) = n(pe^0 + q)^{n-1} \cdot pe^0$

$= n(p + 1 - p)^{n-1} \cdot p$

$\boxed{E(x) = np}$

③ $M_x''(t) = n(n-1)(pe^t + q)^{n-2} \cdot pe^t pe^t + n(pe^t + q)^{n-1} \cdot pe^t$

$M_x''(0) = n(n-1)(p+q)^{n-2} P(1) P(1) + n(p+q)^{n-1} \cdot p$

$= n(n-1)p^2 + np$

$E(x^2) = n^2p^2 - np^2 + np$

∴ ④ $V(x) = E(x^2) - [E(x)]^2$

$= n^2p^2 - np^2 + np - n^2p^2$

$= np(1-p)$

$\boxed{V(x) = npq}$

Binomial Dis ① $P(x) = {}^{n}C_x\, p^x q^{n-x}$

② $\mu = np$

③ $V(x) = npq$

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II) Geometric Distribution

A r·v X is said to have geometric distribution if X assumes non-negative values. i.e $X = 0, 1, 2, 3 \ldots$ } concerned with first success only

$$p.d.f\ /P(x) = \begin{cases} q^x p, & x = 0, 1, 2 \ldots \\ 0, & \text{otherwise} \end{cases} \qquad 0 < p \leq 1, \quad p = 1 - q$$

p → prob success

q → prob. failure

$\text{mean} = \frac{1}{p}$

$\text{variance} = \frac{q}{p^2}$

① $P(x) \geq 0 \quad x = 0, 1, 2$

② $\sum_{k=1}^{\infty} P(x=k) = \sum_{k=1}^{\infty} q^{k-1} p = p + qp + q^2 p \ldots$

$= 1$

$= p(1 + q + q^2 \ldots)$

$= p\left(\frac{1}{1-q}\right)$

$= \frac{p}{p} = 1$

Proof:

① Mean $= \mu = E(x)$

$= \sum x_i P(x_i)$

$= \sum_{k=1}^{\infty} k p q^{k-1}$

$= p \sum_{k=1}^{\infty} k q^{k-1}$

$= p \sum_{k=1}^{\infty} \frac{d}{dq}(q^k)$

$= p \frac{d}{dq}\left(\sum_{k=1}^{\infty} q^k\right)$

$= p \frac{d}{dq}\left(\frac{q}{1-q}\right)$

$= p \frac{(1-q)(1) - q(-1)}{(1-q)^2}$

$= \frac{p(1)}{p^2}$

$\boxed{\mu = \frac{1}{p}}$

② Variance

$E(x^2) = \sum_{k=1}^{\infty} k^2 p q^{k-1}$

$= \sum_{k=1}^{\infty} k(k+1) p q^{k-1} - \sum_{k=1}^{\infty} k q^{k-1} p$

$= p \frac{d}{dq}\left(\sum_{k=1}^{\infty} \left(\frac{d}{dq}\right) q^k\right) - \frac{1}{p}$

$= \frac{2}{p^2} - \frac{1}{p}$

$\therefore V(x) = E(x^2) - E(x)$

$= \frac{2}{p^2} - \frac{1}{p} - \frac{1}{p^2}$

$= \frac{1}{p^2} - \frac{1}{p}$

$= \frac{1}{p}\left(\frac{1}{p} - 1\right)$

$= \frac{1-p}{p^2} = \boxed{\frac{q}{p^2}}$

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III) Poisson Distribution

Let X be a binomial random variable with parameters

$n, p,$ pdf $P(x) = {}^nC_x p^x q^{n-x}$

Then for $n \to \infty$ (trials), $p \to 0$ (prob) $\Big\}$ $np \cong$ constant $= m$ (parameter)

Then the prob. distribution given by the function is called POISSON DISTRIBUTION : $P(x) = e^{-m} \frac{m^x}{x!}$

① $P(x) \geq 0$

② $\sum P(x) = 1$

$m \to$ any parameter.

Proof: Poisson approximation to binomial distribution

As $n \to \infty$, $p \to 0$ $\Big\}$ s.t $np = m$ (m is a +ve real no)

Then $B(x, n, p) \longrightarrow \frac{m^x e^{-m}}{x!}$, $x = 0, 1, 2 \cdots$

$B(x, n, p) = {}^nC_x \, p^x q^{n-x}$

$= {}^nC_x \, p^x (1-p)^{n-x}$ ($np = m$, $\therefore p = m/n$)

$= \frac{n!}{(n-x)!\,x!} \left(\frac{m}{n}\right)^x \left(1 - \frac{m}{n}\right)^{n-x}$

$= \frac{n(n-1)(n-2)\cdots(n-(x+1))(n-x)!}{(n-x)!\,x!} \frac{m^x}{n^x} \frac{\left(1-\frac{m}{n}\right)^n}{\left(1-\frac{m}{n}\right)^x}$

$= \frac{m^x}{x!} \; \frac{n \times n\left(1-\frac{1}{n}\right) \times n\left(1-\frac{2}{n}\right) \cdots n\left(1-\frac{(x+1)}{n}\right)}{n^x} \times \frac{\left(1-\frac{m}{n}\right)^n}{\left(1-\frac{m}{n}\right)^x}$

$= \frac{m^x}{x!} \; \frac{\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{(x+1)}{n}\right)}{} \times \frac{\left(1-m/n\right)^n}{\left(1-m/n\right)^x}$

$\lim_{n\to\infty} \left(1-\frac{m}{n}\right)^n = e^{-m}$

Let $n \to \infty$ and $m = np$

$\lim_{n\to\infty} P(X = x) = \lim_{n\to\infty} \left(\frac{m^x}{x!} \left[\underbrace{\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right) \cdots \left(1-\frac{(x+1)}{n}\right)}_{\to 1} \right] \underbrace{\frac{\left(1-m/n\right)^n}{\left(1-m/n\right)^x}}_{\to 1} \right) = \frac{m^x}{x!} \underbrace{\left(1-\frac{m}{n}\right)^n}_{e^{-m}}$

$= \boxed{\frac{m^x e^{-m}}{x!}}$

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Mean and Variance (from mgf)

$$P(x) = \frac{m^x e^{-m}}{x!}$$

$$M_x(t) = \sum_{x=0}^{\infty} e^{tx} P(x)$$

$$= \sum_{0}^{\infty} e^{tx} \frac{m^x e^{-m}}{x!}$$

$$= e^{-m} \sum_{0}^{\infty} \frac{e^{(tx)} m^x}{x!}$$

$$= e^{-m} \left(1 + \frac{me^t}{1!} + \frac{m^2 e^{2t}}{2!} + \dots\right)$$

$$= e^{-m} \left(1 + \frac{me^t}{1!} + \frac{(me^t)^2}{2!} \dots\right)$$

$$= e^{-m} e^{me^t}$$

$$= e^{me^t - m}$$

$$\boxed{M_x(t) = e^{m(e^t - 1)}} \quad t \in \mathbb{R}$$

Mean:

$$M_x'(t) = e^{m(e^t-1)} \times m(e^t)$$

$$\boxed{E(x)} = M_x'(0) = e^{m(1-1)} \times m(e^0)$$

$$\boxed{\text{Mean} = m}$$

$$M_x''(t) = \left(e^{m(e^t-1)} \cdot me^t\right) me^t + e^{m(e^t-1)} \cdot (me^t)$$

$$E(x^2) = M_x''(0) = \left(e^{m(1-1)} \cdot me^0\right) me^0 + e^{m(e^0-1)} me^0$$

$$= m^2 + m$$

$\therefore$ Variance

$$V(x) = E(x^2) - [E(x)]^2$$

$$= m^2 + m - m^2$$

$$\boxed{V(x) = m}$$

① $P(x) = \frac{m^x e^{-m}}{x!}$

② $\mu = m$

③ $V(x) = m$

$m \to$ any parameter

$\underline{m = np}$

$n \to \infty$

$p \to 0$

* uni parametric

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IV) Uniform Distribution ~~Questions~~

Continuous

Continuous prob. distribution for which prob. density func^n is given by:

$$f(x) = \begin{cases} \frac{1}{b-a}, & a < x < b \\ 0, & \text{otherwise} \end{cases}$$

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_a^b \frac{1}{b-a}\,dx = \frac{x}{b-a}\Big|_a^b = \frac{b}{b-a} - \frac{a}{b-a} = 1$$

m.g.f $\Rightarrow E(e^{tx}) = \int_{-\infty}^{\infty} e^{tx} f(x)\,dx$

$$= \int_a^b e^{tx} \frac{1}{b-a}\,dx$$

$$= \frac{1}{b-a} \int_a^b e^{tx}\,dx$$

$$= \frac{1}{b-a} \frac{e^{tx}}{t}\Big|_a^b$$

$$M_x(t) = \frac{1}{(b-a)} \frac{e^{bt} - e^{at}}{t} \quad t \neq 0$$

$$M_x'(t) = \left(\frac{1}{(b-a)}\right) \frac{t(e^{bt}b - e^{at}a) - (e^{bt} - e^{at})}{t^2}$$

$$= \frac{t((e^{bt})(b-1) - e^{at}(a-1))}{(b-a)t^2}$$

$$M_x'(0) = \lim_{t \to 0} \frac{t(be^{bt} - ae^{at}) - (e^{bt} - e^{at})}{t^2}$$

Using L-H twice.

$$be^{bt} - ae^{at} - (be^{bt} - ae^{at})$$

$$\vdots$$

we get $\mu = \frac{a+b}{2}$.

$$\text{Mean} = \int_{-\infty}^{\infty} x f(x)\,dx$$

$$= \int_a^b \frac{x}{b-a}\,dx$$

$$= \frac{x^2}{2(b-a)}\Big|_a^b$$

$$= \frac{b^2 - a^2}{2(b-a)}$$

$$= \frac{a+b}{2}$$

$$E(x^2) = \int_a^b \frac{x^2}{b-a}\,dx$$

$$= \frac{b^3 - a^3}{3(b-a)}\,dx$$

$$= \frac{(b-a)(b^2+ab+a^2)}{3(b-a)}$$

$$V(x) = E(x^2) - E(x)^2$$

$$= \frac{a^2+ab+b^2}{3} - \frac{a^2+b^2+2ab}{4}$$

$$= \frac{4a^2 - 3a^2 + 4ab - 6ab + 4b^2 - 3b^2}{12}$$

$$= \frac{a^2 - 2ab + b^2}{12} = \boxed{\frac{(a-b)^2}{12}}$$

$$f(x) = \begin{cases} \frac{1}{b-a}, & a < x < b \\ 0, & \text{otherwise} \end{cases}$$

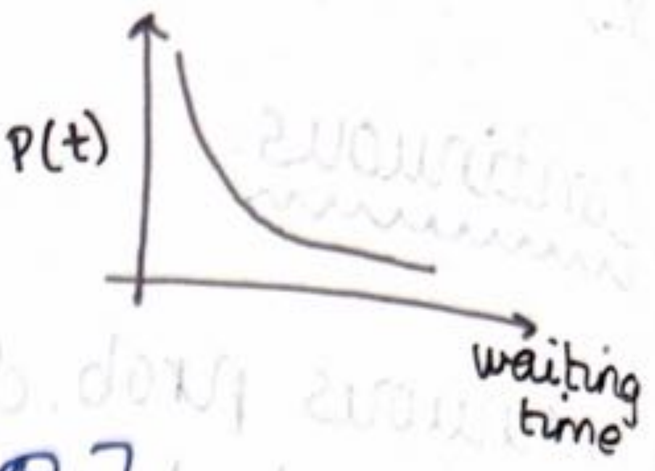
$$\mu = \frac{a+b}{2}, \quad V(x) = \frac{(a-b)^2}{12}.$$

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V) EXPONENTIAL DISTRIBUTION

continuous

[- how long until the next event
  i.e time btwn 2 events



① Probability Distribution $f(x) = \begin{cases} \lambda e^{-\lambda x} & , 0 \le x < \infty \\ 0 & , \text{elsewhere} \end{cases}$

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{0}^{\infty} \lambda e^{-\lambda x}\,dx = \lambda \left.\frac{e^{-\lambda x}}{-\lambda}\right|_{0}^{\infty} = -1\left[\lim_{t\to\infty} e^{-\lambda t} - e^{0}\right] = -[0-1] = 1$$

② C·D·F $F(t) = \int_{0}^{t} f(x)\,dx = \lambda \int_{0}^{t} e^{-\lambda x}\,dx = \lambda\left[\frac{e^{-\lambda x}}{-\lambda}\right]\Big|_{0}^{t}$

$= 1 - e^{-\lambda t}$

③ Mean

$= \int_{0}^{\infty} x\, f(x)\,dx$

$= \int_{0}^{\infty} t\,\lambda e^{-\lambda t}\,dt$

take $\lambda t = x$

$\lambda\,dt = dx$

$= \int_{0}^{\infty} x\, e^{-x}\,\frac{dx}{\lambda}$

$= \frac{1}{\lambda}\int_{0}^{\infty} x\, e^{-x}\,dx$

$(\Gamma(n) = \int_{0}^{\infty} e^{-x} x^{n-1}\,dx$

$\therefore = \frac{1}{\lambda}\Gamma(2)$

$= \boxed{\frac{1}{\lambda}}$

④ Variance

$= E(x^2) - [E(x)]^2$

$= \int_{0}^{\infty} t^2 \lambda e^{-\lambda t}\,dt - \left(\frac{1}{\lambda}\right)^2$

take $\lambda t = x$

$\lambda\,dt = dx$

$= \left[\int_{0}^{\infty} x \times \frac{x}{\lambda}\, e^{-x}\,\frac{dx}{\lambda}\right] - \left(\frac{1}{\lambda}\right)^2$

$= \frac{1}{\lambda^2}\int_{0}^{\infty} x^2 e^{-x}\,dx - \left(\frac{1}{\lambda}\right)^2$

$= \frac{1}{\lambda^2}\,\Gamma(3) - \frac{1}{\lambda^2}$

$= \boxed{\frac{1}{\lambda^2}}$

⑤ mgf

$mgf = E(e^{tx})$

$= \int_{0}^{\infty} e^{tx} f(x)\,dx$

$= \int_{0}^{\infty} e^{tx} \cdot \lambda e^{-\lambda x}\,dx$

$= \lambda \int_{0}^{\infty} e^{-x(\lambda - t)}\,dx$

$= \lambda \left.\frac{e^{-x(\lambda-t)}}{-(\lambda-t)}\right|_{0}^{\infty}$

$= \lambda\left(0 - \frac{e^{0}}{-(\lambda-t)}\right)$

$= \boxed{\frac{\lambda}{\lambda - t}}$

$= \boxed{\frac{1}{\lambda}}$

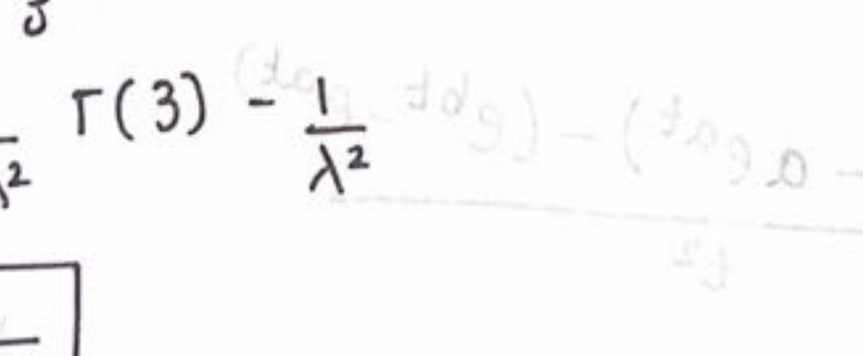
$Pdf = \lambda e^{-\lambda x}\ (0 < x < \infty)$

$Mean = 1/\lambda$

$Variance = 1/\lambda^2$

Note: Memory Loss Property

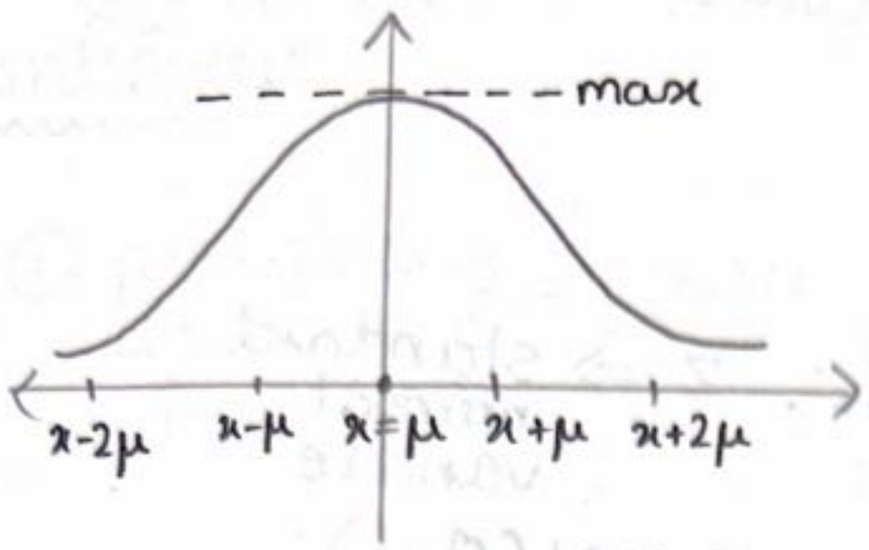
$$P(X \ge s+t \mid X > s) = P(X \ge t)$$



$\therefore P\frac{(X \ge s+t) \cap P(X>s)}{P(X>s)} = P\frac{(X \ge s+t)}{P(X>s)} = P(X \ge t)$

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VII) NORMAL DISTRIBUTION $[X \sim N(\mu, \sigma^2)]$



- highest at the mean
- symmetric about the mean
- $|x-\mu| \to$ larger
  curve $\to$ smaller

The function $e^{-(x-\mu)^2}$ is largest when $\boxed{x=\mu}$

$$f(x) \propto e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

parameters $\Rightarrow \mu$ & σ^2

$$\boxed{f(x) = \frac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad \begin{matrix}\sigma>0 \\ -\infty<x<\infty \\ -\infty<\mu<\infty\end{matrix}}$$

① pdf of normal distribution.

$$\boxed{\text{mgf} = M_x(t) = e^{\left(\frac{\sigma^2 t^2}{2} + t\mu\right)}}$$

② mgf of normal distribution.

③ Mean

$$M_x'(t) = e^{\frac{\sigma^2 t^2}{2} + t\mu} \times \left(\frac{2t\sigma^2}{2} + \mu\right)$$

$$M_x'(0) = E(x) = e^{\frac{\sigma^2(0)}{2} + 0} \times (2\times 0 + \mu)$$

$$\boxed{E(x) = \mu}$$

④ Variance

$$M_x''(t) = M_x(t)\,(\mu + t\sigma^2)$$

$$M_x''(t) = M_x'(t)\,(\mu + t\sigma^2) + M_x(t)\,(*\sigma^2).$$

$$M_x''(0) = E(x^2) = \mu(\mu + 0) + M_x(0)(\sigma^2)$$

$$= \mu^2 + \sigma^2$$

$$\therefore V(x) = E(x^2) - E(x)^2$$

$$= \mu^2 + \sigma^2 - \mu^2$$

$$\boxed{V(x) = \sigma^2}$$

Properties of Normal Distribution.

① curve $\to$ bell shaped, symmetric abt line $x=\mu$

② max. prob. occurs at $x=\mu$
given by $f(x)_{max} = \frac{1}{\sigma\sqrt{2\pi}}$

③ x-axis is the asymptote of the curve.

④ Area:

$\Rightarrow$ If $X \sim N(\mu, \sigma^2)$

Then $P(\mu < X < x_1)$

$$= \int_{\mu}^{x_1} f(x)\,dx = \int_{\mu}^{x_1} \frac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

put $\frac{x-\mu}{\sigma} = z$ $\quad\therefore dx = \sigma\,dz$

$x=\mu,\ z=0$

$x=x_1,\ z=\frac{x_1-\mu}{\sigma}$

$$\therefore \int_0^{\frac{x_1-\mu}{\sigma}} \frac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{z^2}{2}}\, \sigma\,dz$$

$$= \frac{1}{\sqrt{2\pi}} \int_0^{z_1} e^{-z^2/2}\,dz$$

$$= \boxed{\int_0^{z_1} \phi(z)\,dz}$$

$$\phi(z) = \frac{1}{\sqrt{2\pi}}\, e^{-z^2/2}$$

↳ normal prob. integral, gives area under the curve between $z=0$ & $z=z_1$

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⊛ Z (standard normal variate)

⊛ Count = $n \times P(a<X<b)$

Let $X \sim N(\mu, \sigma^2)$

and $Z = \frac{X-\mu}{\sigma}$ } $[X = \mu + \sigma Z]$

$$E(Z) = E\left(\frac{X-\mu}{\sigma}\right) = E\frac{(x)-\mu}{\sigma} = \frac{\mu-\mu}{\sigma} = 0$$

$$V(Z) = V\left(\frac{X-\mu}{\sigma}\right) = \frac{1}{\sigma^2} V(x) = \frac{1}{\sigma^2} \times \sigma^2 = 1$$

$\therefore$ Z ⇒ standard normal variate

$Z \sim N(0, 1)$

⊛ Conversions.

① $P(X \geq a) = P\left(\frac{X-\mu}{\sigma} \geq \frac{a-\mu}{\sigma}\right) = P\left(Z \geq \frac{a-\mu}{\sigma}\right)$

② $P(a \leq x \leq b) = P\left(\frac{a-\mu}{\sigma} \leq \frac{x-\mu}{\sigma} \leq \frac{b-\mu}{\sigma}\right) = P\left(\frac{a-\mu}{\sigma} \leq Z \leq \frac{b-\mu}{\sigma}\right)$

⊛ cdf of standard normal distributio

$\phi(z) \Rightarrow$ i.e

$\phi(z=a) = P(z \leq$

① $\phi(a) = P(z \leq a)$

② $\phi(-a) = 1 - \phi(a)$

③ $P(a<z<b) = \phi(b) - \phi($

⊛⊛ Show that: $E((X-\mu)^{2n}) = 1.3.5 \ldots (2n-1)\sigma^{2n}$

for normal distribution with mean = μ

s.d = σ

var = σ^2

Soln

① mgf of normal dis = $M_x(t) = e^{\frac{\sigma^2 t^2}{2} + t\mu}$

② Let $Y = aX + b$

Then $M_Y(t) = e^{bt} M_x(at)$

③ Compare $X - \mu$ with $aX + b$

$\therefore a = 1, b = -\mu$

$\therefore M_{X-\mu}(t) = e^{-\mu t} M_x(t)$

$= e^{-\mu t} e^{\frac{\sigma^2 t^2}{2} + t\mu}$

$= e^{\frac{\sigma^2 t^2}{2}}$

④ $\therefore M_{x-\mu}(t) = e^{\frac{\sigma^2 t^2}{2}} = 1 + \frac{\sigma^2 t^2}{2(1!)} + \left(\frac{\sigma^2 t^2}{2}\right)^2 \times \frac{1}{2!} + \cdots \left(\frac{\sigma^2 t^2}{2}\right)^n \times \frac{1}{n!}$

⑤ $E((X-\mu)^{2n}) =$ coefficient of $\frac{t^{2n}}{n!}$ in expansion of $M_{(x-\mu)t}$.

$= \frac{\sigma^{2n}(2n)!}{2^n(n!)}$

⑥ $\frac{\sigma^{2n}(2n!)}{2^n(n!)}$

$= \frac{\sigma^{2n}}{2^n} \frac{(1.2.3.4.\ldots.2n)}{(1.2.3.4.\ldots.n)}$

$= \frac{\sigma^{2n}}{2^n} \frac{(1.3.5.7\ldots(2n-1))(2.4.6.8.\ldots.2}{(1.2.3\ldots n)}$

$= \frac{\sigma^{2n}}{2^n} (1.3.5.\ldots(2n-1)) \frac{2^n(1.2.3\ldots n)}{(1.2.3\ldots n)}$

$= \boxed{1.3.5.7\ldots(2n-1) \times \sigma^{2n}}$

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VII) GAMMA DISTRIBUTION

continuous

① probability distribution funcn $= f(x) = \begin{cases} \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} \cdot e^{-x/\beta} & , \ x>0, \ \alpha,\beta>0 \\ 0 & , \ \text{otherwise} \end{cases}$

⊛ Some properties of gamma function

① $\Gamma(n) = \int_0^\infty x^{n-1} \cdot e^{x} \, dx$ ② $\Gamma(n+1) = n\Gamma(n)$ ③ $\Gamma(n+1) = n! \ (n \in \mathbb{N})$ ④ $\Gamma(1/2) = \sqrt{\pi}$

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_0^\infty \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} \cdot e^{-x/\beta} \, dx.$$

Sub $\frac{x}{\beta} = t \Rightarrow dx = \beta\,dt$

$x=0, \ t=0$

$x=\infty, \ t=\infty$

$$= \int_0^\infty \frac{1}{\beta^\alpha \Gamma(\alpha)} (\beta t)^{\alpha-1} e^{-t} \beta \, dt.$$

$$= \int_0^\infty \frac{1}{\beta^\alpha \Gamma(\alpha)} \beta^{\alpha-1} \cdot \beta \, t^{\alpha-1} \cdot e^{-t} \, dt$$

$$= \int_0^\infty \frac{1}{\Gamma(\alpha)} t^{\alpha-1} \cdot e^{-t} \, dt$$

$$= \frac{1}{\Gamma(\alpha)} \times \Gamma(\alpha) = \boxed{1}$$

② Mean $\Rightarrow \mu = \alpha\beta = E(x)$

$$\mu = \int_{-\infty}^{\infty} x f(x)\,dx$$

$$= \int_0^\infty x \frac{1}{\beta^\alpha \Gamma(\alpha)} \cdot x^{\alpha-1} \cdot e^{-x/\beta} \, dx.$$

$$= \frac{1}{\beta^\alpha \Gamma(\alpha)} \int_0^\infty x^{\alpha} \cdot e^{-x/\beta} \, dx$$

sub $\frac{x}{\beta} = t$

$$= \frac{1}{\beta^\alpha \Gamma(\alpha)} \int_0^\infty (\beta t)^{\alpha} \, e^{-t} \beta \, dt.$$

$$= \frac{1}{\Gamma(\alpha)} \int_0^\infty t^{\alpha} \, e^{-t} \, \beta \, dt$$

$$= \frac{\beta}{\Gamma(\alpha)} \Gamma(\alpha+1)$$

$$= \frac{\alpha\beta\,\Gamma(\alpha)}{\Gamma(\alpha)} = \boxed{\alpha\beta}$$

③ Variance $\Rightarrow V(x) = \alpha\beta^2 = \sigma^2$

$$E(x^2) = \int_0^\infty x^2 f(x)\,dx$$

$$= \int_0^\infty x^2 \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} \cdot e^{-x/\beta} \, dx$$

$$= \frac{1}{\beta^\alpha \Gamma(\alpha)} \int_0^\infty x^{\alpha+1} \cdot e^{-x/\beta} \, dx$$

sub $\frac{x}{\beta} = t$.

$$= \frac{1}{\beta^\alpha \Gamma(\alpha)} \int_0^\infty (\beta t)^{\alpha+1} e^{-t} \beta \, dt.$$

$$= \frac{1}{\beta^\alpha \Gamma(\alpha)} \int_0^\infty \beta^\alpha \cdot \beta^2 \cdot t^{\alpha+1} e^{-t} \, dt$$

$$= \frac{\beta^2}{\Gamma(\alpha)} \int_0^\infty t^{\alpha+1} e^{-t} \, dt.$$

$$= \frac{\beta^2}{\Gamma(\alpha)} \Gamma(\alpha+2) = (\alpha+1)\alpha\beta^2$$

$\therefore \boxed{V(x)} = E(x^2) - [E(x)]^2$

$= \alpha^2\beta^2 + \alpha\beta^2 - \alpha^2\beta^2 = \boxed{\alpha\beta^2}$

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VIII) IMPORTANT INEQUALITIES

MARKOV'S INEQUALITY

For any non-negative r.v X i.e $(X \geq 0)$ and $(a > 0)$ (+ve)

$$P(X \geq a) \leq \frac{E(X)}{a}$$

Proof: $X \geq a$

$E(X) \geq E(a \cdot 1_{x \geq a})$

$E(x) \geq a\ P(x \geq a)$

$\therefore P(x \geq a) \geq \frac{E(x)}{a}$

CHEBYSHEV'S INEQUALITY

Let X: r.v with finite mean μ & variance σ^2

Then for any $\varepsilon > 0$, $P\{|x-\mu| \geq \varepsilon\} \leq \frac{\sigma^2}{\varepsilon^2}$

Proof: ① $(x-\mu)^2 \geq 0$ [non-negative random variable]

② Take $a = \varepsilon^2 > 0$ in Markov's Inequality

$$P\{(x-\mu)^2 \geq a^2\} \leq \frac{E(X-\mu)^2}{a^2}$$

$$P\{(x-\mu)^2 \geq \varepsilon^2\} \leq \frac{\sigma^2}{\varepsilon^2}$$

$\therefore P\{|X-\mu| \geq \varepsilon\} \leq \frac{\sigma^2}{\varepsilon^2}$ proved

NOTE: ⚠⚠

① $P(|x-\mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}$ (upper bound)

② $P(|x-\mu| \geq k\sigma) \leq \frac{1}{k^2}$ (upper bound)

③ $P(|x-\mu| < k\sigma) \geq 1 - \frac{1}{k^2}$ (lower bound)

suppose $k\sigma = \varepsilon$

$\frac{\sigma^2}{\varepsilon^2} = \frac{1}{k^2}$

important properties [mgf's]

Property 1: Suppose x and Y are independent variables

Let $Z = X+Y$

$M_x(t), M_y(t), M_z(t)$ are mgf's of X, Y, Z

$\therefore M_z(t) = M_x(t) \cdot M_y(t)$

Generalisation: $M_z(t) = M_{x_1}(t) \cdot M_{x_2}(t) \cdots M_{x_n}(t)$ $1 \leq x_i \leq n$ (x_i = random variable)

Property 2: If $X_i \sim N(\mu, \sigma^2)$ $[1 \leq i \leq n]$ [x_i: 'n' independent variables]

Then $\Rightarrow Z = X_1 + X_2 + \cdots X_n$

$$Z = N\left(\sum_1^n \mu_i, \sum_1^n \sigma_i^2\right)$$

Property 3: $Z = X - Y$

$$M_z(t) = e^{(\mu_1-\mu_2)t + \left(\frac{\sigma_1^2+\sigma_2^2}{2}\right)t^2}$$

$Z \sim N(\mu_1-\mu_2, \sigma_1^2+\sigma_2^2)$

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#Note : Let E be an experiment associated with a sample space S

Let X be a random variable

R_x be its range space

Suppose B is an event, $B \subseteq R_x$

Then A is an event defined as: $A = \{ s \in S_s \mid X(s) \in B \}$

[A is equivalent to B].

#Note : ① Given:

$X \rightarrow$ crv

pdf of $x \rightarrow f(x)$

$Y = H(x)$

pdf of $y = g(y) = ??$

① obtain $G(y) = P(Y \leq y)$

② differentiate $G(y)$ w.r.t y to get $g(y)$

③ determine values of y, in range space of Y for which $g(y) > 0$

② p.d.f of $y = g(y) = f(x) \left| \frac{dx}{dy} \right|$

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X) LAW OF LARGE NUMBERS

Let $x_1, x_2, x_3, \ldots, x_n$ be a sequence; independent and identically distributed random variables, each having finite mean $E(x_i) = \mu$ [finite]

Then for any $\varepsilon > 0$ $\quad P\left\{ \left| \frac{x_1 + x_2 + \ldots x_n}{n} - \mu \right| \geq \varepsilon \right\} \to 0$

(i.e the probability that the sample average deviates from the mean μ by at least ε goes to zero as the no. of samples n becomes very large) as $n \to \infty$

Proof:

① $E\left(\frac{x_1 + x_2 + \ldots + x_n}{n}\right)$

$= \frac{E(x_1) + E(x_2) + \ldots E(x_n)}{n}$

$= \frac{n\mu}{n} = \mu$

② $V\left(\frac{x_1 + x_2 + x_3 + \ldots + x_n}{n}\right)$

$= \frac{1}{n^2}\left(V(x_1) + V(x_2) + \ldots V(x_n)\right)$

$= \frac{1}{n^2}(n\sigma^2) = \frac{\sigma^2}{n}$

③ Using Chebyshev's Inequality,

$$P\left\{ \left| \frac{x_1 + x_2 + \ldots + x_n}{n} - \mu \right| \geq \varepsilon \right\} \leq \frac{\sigma^2}{n\varepsilon^2} \quad \begin{matrix} \to 0 \\ \text{as } n \to \infty \end{matrix}$$

XI) CENTRAL LIMIT THEOREM

Let $X_1, X_2, \ldots, X_n$ denote the items of a random sample from a distribution with mean μ and variance σ^2.

Then the random variable

$$Y_n = \frac{\sum_{i=1}^{n} X_i - n\mu}{\sigma\sqrt{n}} \sim N(0, 1)$$ std. normal distribution

$$Z = \frac{\sum_{i=1}^{n} \frac{X_i}{n} - \frac{n\mu}{n}}{\frac{\sigma\sqrt{n}}{n}} = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

$\left[Y_n = \frac{S - E(S)}{V(S)} \right]$

Proof (v.v. imp)

① calculate mean, variance and Y_n for S_n

Let $S_n = X_1 + X_2 + X_3 + \ldots + X_n$

i) $E(S_n) = E(X_1 + X_2 + \ldots + X_n)$

$= E(X_1) + E(X_2) + \ldots E(X_n)$

$\therefore E(S_n) = n\mu$

(ii) $V(S_n) = V(X_1 + X_2 + \ldots + X_n)$

$= V(X_1) + V(X_2) + \ldots + V(X_n)$

$= n\sigma^2$

(iii) $Y_n = \frac{\sum X_i - n\mu}{\sigma\sqrt{n}} = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$

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2) calculate mgf of Y_n

mgf of Y_n

$$= M_{Y_n}(t) = E\left(e^{tY_n}\right)$$

$$= E\left(e^{t\frac{(\sum x_i - n\mu)}{\sigma\sqrt{n}}}\right)$$

$$= E\left(e^{t\frac{(x_1-n\mu)}{\sigma\sqrt{n}} + t\frac{(x_2-\mu)}{\sigma\sqrt{n}} + \cdots t\frac{(x_n-\mu)}{\sigma\sqrt{n}}}\right)$$

$$= E\left(e^{t\frac{(x_1-\mu)}{\sigma\sqrt{n}}} \cdot e^{t\frac{(x_2-\mu)}{\sigma\sqrt{n}}} \cdot \cdots \cdot e^{t\frac{(x_n-\mu)}{\sigma\sqrt{n}}}\right)$$

$$= E\left(\prod_{i=1}^{n} e^{t\frac{(x_i-\mu)}{\sigma\sqrt{n}}}\right) \quad [\because x_i \text{ is independent}]$$

$$M_{Y_n}(t) = \left[E\left(e^{t\frac{(x_i-\mu)}{\sigma\sqrt{n}}}\right)\right]^n \quad [\because \text{random var are identically distributed over } E(x_i)=\mu \text{ and } V(x_i)=\sigma^2]$$

$$= \left[E\left(1 + t\frac{(x_i-\mu)}{\sigma\sqrt{n}} + \frac{t^2(x_i-\mu)^2}{\sigma^2 n(2!)} + O(n^{-3/2})\right)\right]^n$$

$(\because (n^{-1/2})^3 = O(n^{-3/2}))$

$$= \left[1 + \frac{t}{\sigma\sqrt{n}}E(x_i-\mu) + \frac{t^2}{2\sigma^2 n}E(x_i-\mu)^2 + O(n^{-3/2})\right]^n \quad \left(e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \ldots\right)$$

$$= \left[1 + \frac{t}{\sigma\sqrt{n}}(\underbrace{E(x_i)}_{\mu} - \mu) + \frac{t^2}{2\sigma^2 n}E(x_i-\mu)^2\right]^n$$

$$= \left[1 + 0 + \frac{t^2}{2\sigma^2 n}(\sigma^2)\right]^n$$

For any fixed t, terms of $O(n^{-3/2}) \to 0$ as $n \to \infty$

$$\therefore \lim_{n\to\infty} M_{Y_n}(t) = \left[1 + 0 + \frac{t^2}{2n} + O(n^{-3/2})\right]^n$$

$$= \left[1 + \frac{t^2}{2n}\right]^n \quad \left[\because \lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n = e^x\right]$$

$$= e^{t^2/2}$$

$$\therefore e^{t^2/2} \sim N(0,1)$$

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(XII) CHI-SQUARE DISTRIBUTION

Let $Z \sim N(0,1) \Rightarrow$ pdf of $Z = f_z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$

Let $X = Z^2$
$\therefore Z = \pm X$

p.d.f of X

$= f_x(z) = f_z(\sqrt{x}) \left|\frac{dZ}{dX}\right| + f_z(-\sqrt{x}) \left|\frac{dZ}{dx}\right|$

$= \frac{1}{\sqrt{2\pi}} e^{-x/2} \frac{1}{2\sqrt{x}} + \frac{1}{\sqrt{2\pi}} e^{-x/2} \frac{1}{2\sqrt{x}}$

$= \frac{1}{\sqrt{2\pi}} e^{-x/2} \frac{1}{\sqrt{x}}$

$\therefore$ Chi-Square
① PDF (1 degree of freedom) $\Big\}$ $X \sim \chi^2(1)$

$$\boxed{f_x(x) = \frac{1}{\sqrt{2\pi x}} e^{-x/2}, \quad x>0}$$

* If $Z_1, Z_2, Z_3, \ldots Z_k \sim N(0,1)$ independent random variables

Then $X = Z_1^2 + Z_2^2 + \ldots + Z_k^2$ follows chi-square distribution with 'k' degrees of freedom i.e $\chi^2(k)$

[sum of squares of k-independent std normal variates]

$X \sim \chi^2(k) \iff X \sim \text{Gamma}\left(\alpha = \frac{k}{2}, \beta = 2\right)$

From gamma distribution,

$f(x) = \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} \cdot e^{-x/\beta} \quad x>0$ $\xrightarrow[\beta=2]{\text{take } \alpha=k/2}$ $\boxed{f(x) = \frac{1}{\Gamma(k/2)\, 2^{k/2}} x^{k/2-1} e^{-x/2}}$

② pdf of chi-sq. distribution.

③ ~~pdf~~ mgf of χ^2 distribution

$M_x(t) = (1-2t)^{-k/2}, \quad t < 1/2$

④ Mean $\Rightarrow M_x'(t) = -\frac{k}{2}(1-2t)^{-\frac{k}{2}-1}(-2)$

$M_x'(0) = E(x) = k$

$\boxed{E(x) = k}$

⑤ Variance

$M_x''(t) = -\frac{k}{2}\left(-\frac{k}{2}-1\right)(1-2t)^{-\frac{k}{2}-2}(-2)$

$E(x^2) = M_x''(0) = k(k+2)$

$V(x) = E(x^2) - E(x)^2$
$= k(k+2) - k^2$

$\boxed{V(x) = 2k}$

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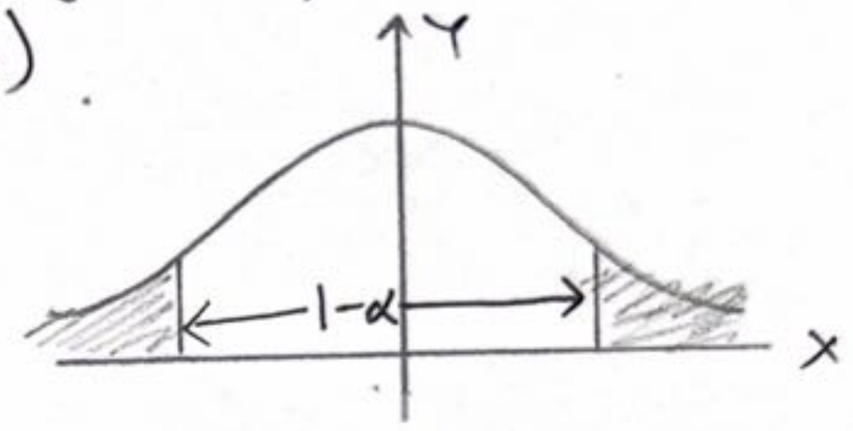
VI) INTERVAL ESTIMATION : CONFIDENCE INTERVAL.

An interval with one or both of its endpoints as random variables is called a random interval. eg: $(X, 2X)$, $(2, \bar{X})$.

Confidence Interval :- $\theta \Rightarrow$ unknown parameter
$(X_1, X_2 \ldots X_n) \Rightarrow$ random sample of size n

The confidence interval for the parameter θ is a random interval containing the parameter with high probability, say $(1-\alpha)$

$1-\alpha \Rightarrow$ Confidence Coefficient.



Note ☆☆

Let $X_1, X_2, \ldots X_n$ be a random sample from the distribution $N(\mu, \sigma^2)$. Then:

① $Z = \dfrac{(\bar{X}-\mu)}{\sigma/\sqrt{n}} \sim N(0,1)$

② $V = \dfrac{nS^2}{\sigma^2} \sim \chi^2(n-1)$

③ $\sqrt{n-1}\ \dfrac{(\bar{X}-\mu)}{S} \sim T(n-1)$

④ $\dfrac{\sum_{i=1}^{n}(X_i-\mu)^2}{\sigma^2} \sim \chi^2(n)$

Confidence Interval for Mean (when σ^2 is known)

$P(-a < Z < a) = 1-\alpha$

[consider $\bar{X} \sim N(\mu, \sigma^2)$

$Z = \dfrac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$]

$\phi(a) - \phi(-a) = 1-\alpha$

$2\phi(a) - 1 = 1-\alpha$

$\boxed{\phi(a) = 1 - \dfrac{\alpha}{2}}$ find 'a' from table

$\left[\bar{X} - \dfrac{a\sigma}{\sqrt{n}}, \bar{X} + \dfrac{a\sigma}{\sqrt{n}}\right]$ is a $(1-\alpha)100\%$ confidence interval for μ, when σ is known.

⊛ Confidence interval for mean (when σ^2 is not known)

Consider, $T = \dfrac{(\bar{X}-\mu)\sqrt{n-1}}{S} \sim T(n-1)$

$P(-a < T < a) = 1-\alpha$

$P\left(-a < \dfrac{(\bar{X}-\mu)\sqrt{n-1}}{S} < a\right) = 1-\alpha.$

$P\left(-\dfrac{aS}{\sqrt{n-1}} < \bar{X}-\mu < \dfrac{aS}{\sqrt{n-1}}\right) = 1-\alpha.$

$P\left(\bar{X} - \dfrac{aS}{\sqrt{n-1}} < \mu < \bar{X} + \dfrac{aS}{\sqrt{n-1}}\right) = 1-\alpha$

$\left(\bar{X} - \dfrac{aS}{\sqrt{n-1}}, \bar{X} + \dfrac{aS}{\sqrt{n-1}}\right)$ is a confidence interval $(1-\alpha)100\%$ when σ is unknown

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5/11/2025

Probability Space : (Ω, G, P)

$\Omega \Rightarrow$ sample space

$G \Rightarrow$ all subsets of Ω

$P \Rightarrow$ prob. measure $P : F \to [0,1]$

if $A_1, A_2, \ldots A_\infty$ in F,

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

eg: prob space for rolling a die

$\Omega = \{1,2,\ldots 6\}$

$A = \{$all even nos$\}$ $\quad P(A) = \frac{3}{6}$

$\in F$

(Ω, F, P) is a prob. space.

Defn:

Let (Ω, F, P) be a prob. space

T: index set (usually time)

S: ~~sss~~ state space (usually subset $\mathbb{R}^n$)

n-tuple of vectors $(x_1, x_2, \ldots x_n)$

$\{X_t : t \in T\}$ such that

for each fixed $t \in T$

$\boxed{X_t : \Omega \to S}$ is measurable function (i.e prob exists)

index set.

$\{X_n : n = 1,2,3\ldots\}$ is called a stochastic process

eg: toss a fair coin several times

H: 1 , T: 0

$\Omega_{\text{sample space}} = \{\infty$ set of 0's & 1's$\}$

$X_n = \begin{cases} 1 & \text{if } n^{th} \text{ toss is head} \\ 0 & \text{if } n^{th} \text{ toss is tail} \end{cases}$ (n^{th} toss).

for $n = 1,2,3\ldots$ $\quad S = \{0,1\}$

$\{X_n\}_{n \geq 1}$

$\{X_n : n = 1,2,\ldots\}$ is called a stochastic process.

Consider the stochastic process

$\{X(t) : t = 0,1,2,\ldots\}$

$X_1 = 1, X_3 = 1 \ldots$

$X_0, X_1, X_2 \ldots$

$X_0 = x_{01}, x_{02}, x_{03} \ldots$ values of X_0

$X_1 = x_{11}, x_{12}, x_{13} \ldots$ values of X_1

$\vdots$

$S = \{x_{ij} \mid \begin{matrix} i = 0,1,2,3\ldots \\ j = 1,2,3\ldots \end{matrix}\}$ state space

① The values taken by X_1 depends on X_0

② The values taken by X_2 depends only on X_1 but not on X_0

③ The values taken by X_3 depends only on X_2 but not on X_1, X_0.

Suppose that for each pair of states (a_i, a_j) and for any successive random variable X_s and X_{s+1} in the process, the conditional prob that the event $X_{s+1} = a_j$ occurs given that $X_s = a_i$ already occurred.

$P(X_{s+1} = a_j \mid X_s = a_i)$

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Suppose the stochastic process

$\{X(t) \mid t=0,1,2\ldots\}$

is called Markov Chain if

for any $j, k, j_1, j_2 \ldots j_{n-1} \in \mathbb{N}$

then $P(X_n=k \mid X_{n-1}=j,\ X_{n-2}=j_1,\ X_{n-3}=j_2 \ldots\ X_0=j_{n-1})$

$= P\{X_n=k \mid X_{n-1}=j\} = P_{jk}\}$

Transition Probabilities

to col ↓

$$\text{from row} \rightarrow \begin{pmatrix} P_{11} & P_{12} & \cdots & P_{1m} \\ P_{21} & P_{22} & \cdots & P_{2m} \\ \vdots & & & \vdots \\ P_{m1} & P_{m2} & \cdots & P_{mm} \end{pmatrix}$$

(1) A student study habits is as follows:

i) if he studies one night, he is 30% sure to study next night.

ii) if he does not study one night he is 40% sure to study next night

Find transition matrix of the chain and his study / Also draw diagram.

① States ⇒ a_1 : studying

a_2 : not studying.

② Transition matrix ⇒

$$\begin{array}{c} \\ a_1 \\ a_2 \end{array} \begin{array}{c} \begin{array}{cc} a_1 & a_2 \end{array} \\ \begin{pmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{pmatrix} \end{array}$$

P_{11}: P(he will study one night | he has studied the previous night)

$= 0.3$

$\therefore P_{12} = 1 - P_{11} = 0.7$

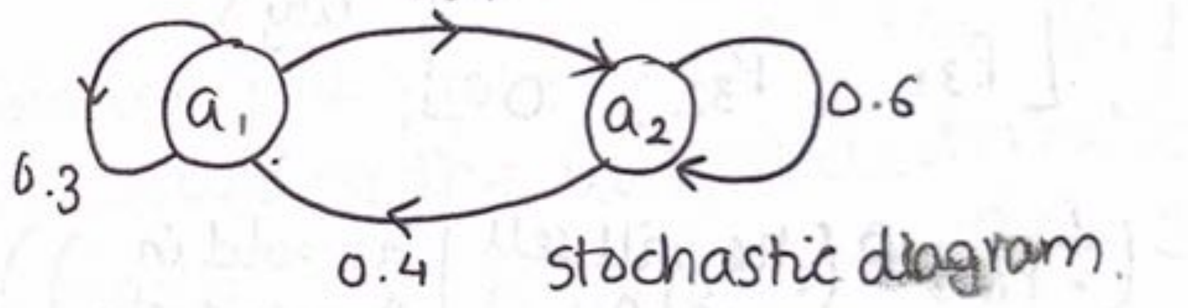
P_{21} = P(he will study one night | not studied previously)

$= 0.4$

$P_{22} = 1 - 0.4 = 0.6$.

$$\begin{array}{c} \\ a_1 \\ a_2 \end{array} \begin{array}{c} \begin{array}{cc} a_1 & a_2 \end{array} \\ \begin{pmatrix} P_{11}=0.3 & P_{12}=0.7 \\ P_{21}=0.4 & P_{22}=0.6 \end{pmatrix} \end{array} \begin{array}{l} \\ \\ \rightarrow \text{sum}=1 \\ \rightarrow \text{sum}=1. \end{array}$$

③ Diagram



stochastic diagram.

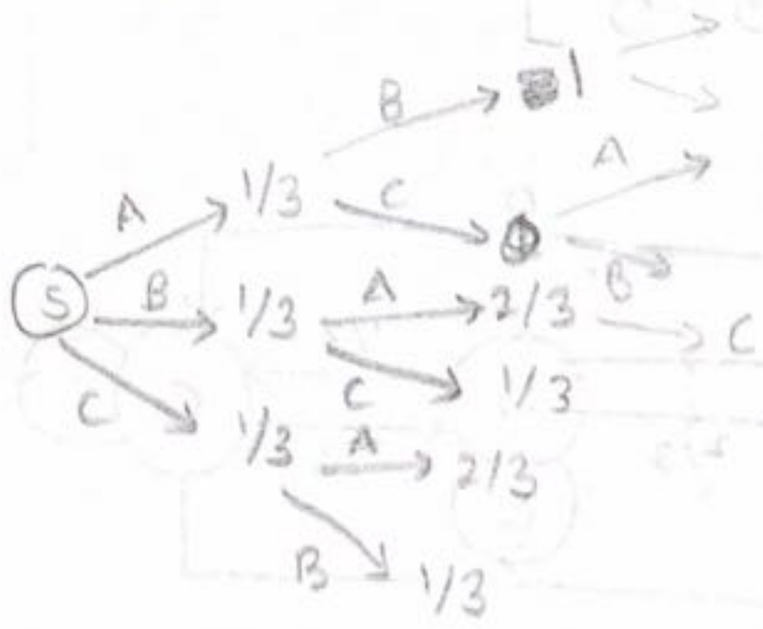
(2) salesman S sells in 3 cities A, B, C.

① He never sells in the same city on successive days.

② If sells in A, then next in city B

③ If he sells in B or C, then next day he is twice as likely to sell in A as in other city

Find transition matrix & state diagram of markov chain.



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Soln:

$P(\text{selling in any city}) = \frac{1}{3}$.

a_1: selling in city A
a_2: " " " B
a_3: " " " C

Transition Matrix

$$\begin{array}{c} \\ a_1 \\ a_2 \\ a_3 \end{array} \begin{array}{c} \begin{array}{ccc} a_1 & a_2 & a_3 \end{array} \\ \begin{bmatrix} P_{11} & P_{12} & P_{13} \\ P_{21} & P_{22} & P_{23} \\ P_{31} & P_{32} & P_{33} \end{bmatrix} \end{array}$$

$$= \begin{bmatrix} 0 & P_{12} & P_{13} \\ P_{21} & 0 & P_{23} \\ P_{31} & P_{32} & 0 \end{bmatrix}$$ (never sells in same city).

$\left(\because P_{11} = P\left(\begin{array}{c|c} \text{he will sell in A} & \text{he sold in A previously} \end{array}\right)\right)$

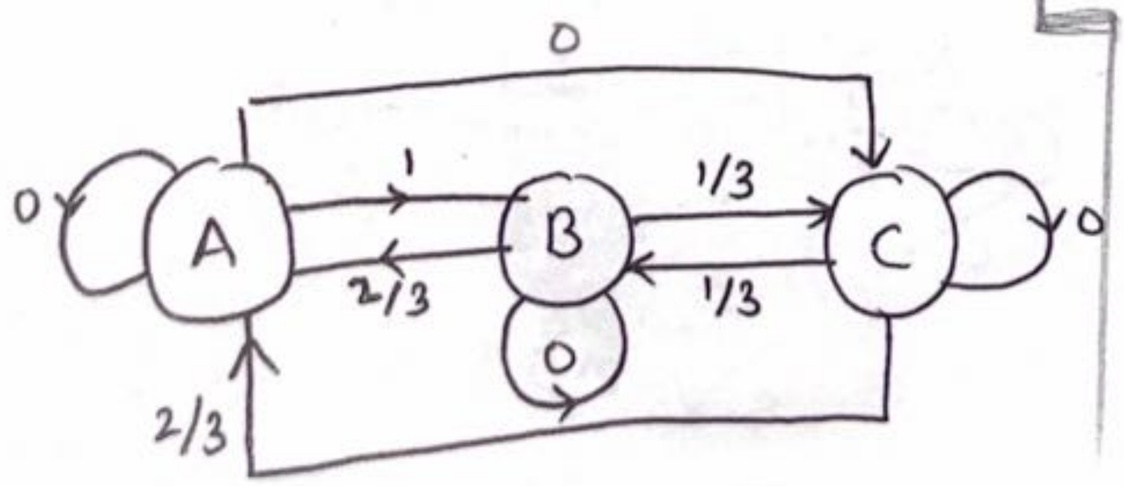
$= 0$

$P_{12} = P\left(\begin{array}{c|c} \text{he will sell in B} & \text{he sold in A prev} \end{array}\right)$

$= 1$

$\therefore P_{13} = 1 - P_{12} = 0$

$$= \begin{bmatrix} 0 & 1 & 0 \\ 2/3 & 0 & 1/3 \\ 2/3 & 1/3 & 0 \end{bmatrix}$$



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,bability Distribution in Markov

Consider Markov Chain $\{x(t) \mid t \in T\}$ with the state space $S = \{a, b, c \ldots i, j, k\}$

Then the prob. distribution of $X_\gamma, X_{\gamma+1}, \ldots X_{\gamma+n}$ can be determined in terms of the transition probabilities P_{jk} and the initial distribution X_γ ($\gamma = 0$, we get X_0)

$$\boxed{P(A \cap B) = P(A|B) \cdot P(B)}$$

Take $\gamma = 0$ $\quad X_0, X_1, \ldots X_n$

The joint prob. of ~~$(X_0 = a, X_1 = b,$~~ $(X_0, X_1, \ldots X_n)$

$= P\{X_0 = a, X_1 = b, X_2 = c \ldots, X_{n-2} = i, X_{n-1} = j, X_n = k\}$

$= P\{\underbrace{X_n = k}_{A}, \underbrace{X_{n-1} = j, X_{n-2} = i \ldots, X_1 = b, X_0 = a}_{B}\}$

$= P\{X_n = k \mid X_{n-1} = j, X_{n-2} = i, \ldots X_1 = b, X_0 = a\}$
$\quad \times P(X_{n-1} = j, X_{n-2} = i, \ldots X_1 = b, X_0 = a\}$

$= P\{X_n = k \mid X_{n-1} = j\} \cdot P\{X_{n-1} = j, X_{n-2} = i, \ldots X_1 = b, X_0 = a\}$

$= P_{jk} \cdot P\{X_{n-1} = j \mid X_{n-2} = i, X_{n-3} = h, \ldots X_1 = b, X_0 = a\}$

$= P_{jk} \cdot P\{X_{n-1} = j \mid X_{n-2} = i, \ldots X_2 = c, X_1 = b, X_0 = a\}$

$= P_{jk} \cdot P_{ij} \{X_{n-2} = i, \ldots X_2 = c, X_1 = b, X_0 = a\}$

$\vdots$ recurrence

$= \boxed{P_{jk} \cdot P_{ij} \cdot P_{hi} \ldots P_{ab} \cdot P\{X_0 = a\}}$

In general,

$P(X_\gamma = a, X_{\gamma+1} = b, \ldots X_{\gamma+n-1} = j, X_{\gamma+n} = k\}$

$= P\{X_\gamma = a \mid P_{ab} \cdot P_{bc} \ldots P_{ij} \cdot P_{jk}\}$

9) Let $\{X_n \mid n \geq 0\}$ be a Markov Chain with 3 states 0, 1, 2 and transition matrix given by $P = \begin{bmatrix} 3/4 & 1/4 & 0 \\ 1/4 & 1/2 & 1/4 \\ 0 & 3/4 & 1/4 \end{bmatrix}$

Find the foll. prob:
by assuming the initial distribution is equally likely to happen for 3 states
i.e $P(X_0 = i) = 1/3$ for $i = 0, 1, 2$.

① $P\{X_1 = 1 \mid X_0 = 2\}$
② $P\{X_2 = 2 \mid X_1 = 1\}$
③ $P\{X_2 = 2, X_1 = 1, X_0 = 2\}$
④ $P\{X_3 = 1, X_2 = 2, X_1 = 1, X_0 = 2\}$.

Draw stochastic diagram.

prev \ present next	0	1	2
0	3/4	1/4	0
1	1/4	1/2	1/4
2	0	3/4	1/4

① $P_{21} = P[X_1 = 1 \mid X_0 = 2] = 3/4$

② $P_{12} = P[X_2 = 2 \mid X_1 = 1] = 1/4$

③ $P(\underbrace{X_2 = 2}_{A}, \underbrace{X_1 = 1, X_0 = 2}_{B})$

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$P(X_2=2, X_1=1, X_0=2\}$

$= P(X_2=2 \mid X_1=1, X_0=2) \cdot P(X_1=1, X_0=2)$

$= P(X_2=2 \mid X_1=1] \cdot P[X_1=1 \mid X_0=2] \cdot P(X_0=2)$

$= P_{12} \cdot P_{21} \cdot P(X_0=2)$

$= 1/4 \cdot 3/4 \cdot \times 1/3$

$= \boxed{1/16}$

④ $P(X_3=1, X_2=2, X_1=1, X_0=2)$.

$= P_{21} \cdot P_{12} \cdot P_{21} \, P(X_0=2)$.

$= \frac{3}{4} \cdot \frac{1}{4} \times \frac{3}{4} \times \frac{1}{3} = \boxed{\frac{3}{64}}$

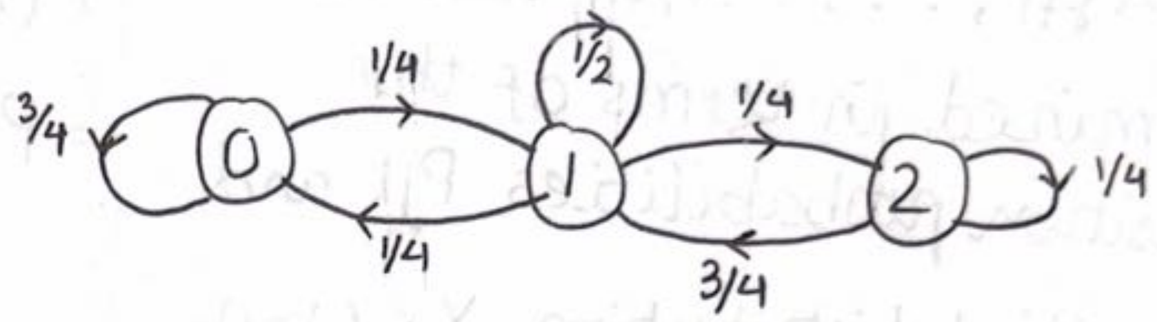
⑤ $P(X_2=2, X_1=1 \mid X_0=2)$

$= P(X_2=2 \mid X_1=1] \, P[X_1=1 \mid X_0=2$

$= P_{22} \cdot P_{21}$

$= \frac{1}{4} \times \frac{3}{4} = \frac{3}{16}$.

Stochastic Diagram



$\boxed{P(A \cap B \mid C)}$

$= \dfrac{P(A \cap B \cap C)}{P(C)}$

$= \dfrac{P(A \mid B) \cdot P(B \mid C) \cdot P(C)}{P(C)}$ ← Markov Chain

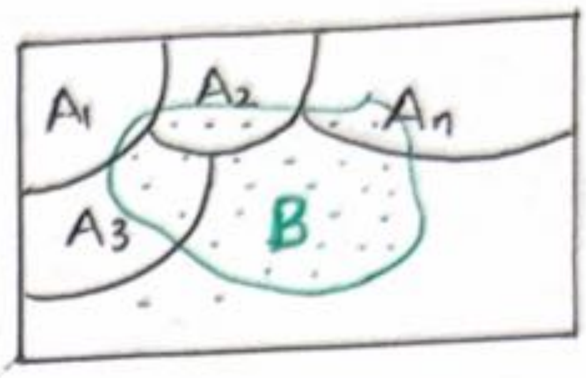
$= \boxed{P(A \mid B) \cdot P(B \mid C)}$

...yes Theorem (Proof!)

Given $A_1, A_2, \ldots A_n$ events

$A_i \cap A_j = \phi$ } partition
$\bigcup_{i=1}^{n} A_i = S$ }

→ mutually exclusive (pairwise)
→ union is sample space.



Let B : event, $B > 0$

By defn of conditional prob
(for any k ⇒ $(1 \leq k \leq n)$)

$$P(A_k \mid B) = \frac{P(A_k \cap B)}{P(B)} \quad \text{—— ①}$$

$$P(B \mid A_k) = \frac{P(B \cap A_k)}{P(A_k)} \quad \text{—— ②}$$

② ⇒ $P(A_k \cap B) = P(B \mid A_k) \cdot P(A_k)$

sub in ①, we get

$$P(A_k \mid B) = \frac{P(B \mid A_k) \cdot P(A_k)}{P(B)} \quad \text{—— ③}$$

disjoint as A_i disjoint.

$B = (A_1 \cap B) \cup (A_2 \cap B) \cup (A_3 \cap B) \ldots (A_n \cap B)$

$P(B) = P(A_1 \cap B) + P(A_2 \cap B) + \ldots P(A_n \cap B)$

$$P(B) = \sum_{i=1}^{n} (A_i \cap B) \quad \text{—— ④}$$

sub ② in ④

$$P(B) = \sum_{i=1}^{n} P(B \mid A_i) \cdot P(A_i) \quad \text{—— ⑤}$$

sub ⑤ in ③, $P(A_k \mid B) = \dfrac{P(B \mid A_k) \cdot P(A_k)}{\sum_{i=1}^{n} P(B \mid A_i) \cdot P(A_i)}$

∴ Finally,

$$P(A_k \mid B) = \frac{P(B \mid A_k) \cdot P(A_k)}{\sum_{i=1}^{n} P(B \mid A_i) \cdot P(A_i)}$$

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Basic Set Formulas.

① $P(\phi) = 0$

② $P(A) \leq P(B)$ (if $A \subseteq B$)

③ $P(A) \leq P(B')$ (if $A \cap B = \phi$)

④ $P(A) + P(\bar{A}) = 1$

⑤ $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

⑥ $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$

⑦ $P(A \cap B') = P(A) - P(B)$ if $B \subseteq A$.

⑧ if A & B independent events
$P(A \cap B) = P(A)\,P(B)$.

⑨ if A, B independent,
$\bar{A}$, $\bar{B}$ also independent.

⑩ $P(E) = \dfrac{\text{no. of fav. outcomes}}{\text{total no. of outcomes}}$

Conditional Probability

$$P(B|A) = \frac{P(B \cap A)}{P(A)}, \quad P(A) \neq 0$$

⇓

P(event B, given A)

Law of Total Probability

$$P(E) = P(A_1) \cdot P(E|A_1) + P(A_2) \cdot P(E|A_2) + \ldots P(A_n) \cdot P(E|A_n)$$

Baye's Theorem

Event F, sample space S, $P(F) > 0$

$\{E_1, E_2, \ldots E_k\} \Rightarrow$ partition of S corr. random experiment

$$P(E_j|F) = \frac{P(F|E_j)\,P(E_j)}{P(E_1)\,P(F|E_1) + P(E_2)\,P(F|E_2) + \ldots P(E_k)\,P(F|E_k)}$$

① c-d-f $f(x) = \sum P_i$

② $E(X) = \mu = \sum_{i=1}^{\infty} x_i P(x_i)$

③ $E(x-\mu)^2 = \sum_{i=1}^{\infty} (x_i - \mu)^2 P(x_i)$ $(V(x))$

① $P(a \leq x \leq b) = \int_a^b f(x)\,dx$.

$P(X \geq r) = \int_r^{\infty} f(x)\,dx$.

② $E(x) = \int x f(x)\,dx$.

③ $V(x) = E(x^2) - [E(x)]^2$.

c·d·f $\int_{-\infty}^{\infty} f(x)\,dx = 1 = \sum_{i=1}^{\infty} P(x_i)$.

① $E(X) = c$, if $P(X = c) = 1$

② $E(aX + b) = a\,E(X) + b$

③ $E(X) \geq 0$

④ $E(X+Y) = E(X) + E(Y)$

⑤ $V(x) = E(x^2) - [E(x)]^2$.
$= E(x-\mu)^2$.

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2 dim random variable

Discrete.

① $P(x_i, y_i) = P(X=x_i, Y=y_j)$

② $P(x_i, y_j) \geq 0$

③ $\sum_{j=1}^{\infty} \sum_{i=1}^{\infty} P(x_i, y_j) = 1$

④ $P(x_i) = \sum_{j=1}^{\infty} P(x_i, y_j)$

Marginals

$Q(y_i) = \sum_{i=1}^{\infty} P(x_i, y_j)$

⑤ Conditional Densities

$P(x_i|y_i) = P(X=x_i \mid Y=y_i)$

$= \frac{P(X=x_i, Y=y_i)}{P(Y=y_i)}$

mgf: $M_x(t) = \sum_{j=1}^{\infty} e^{tx} p(x_j)$

$= E(e^{tx})$

$M_x'(0) = \mu = E(x)$

$M_x''(0) = E(x^2)$

$\vdots$

$M_x^n(0) = E(x^n)$ nth moment.

* $\lim_{n\to\infty} \left(1 - \frac{m}{n}\right)^n = e^{-m}$

* $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$

① $Cov(X, Y) = E[(X - E(X))(Y - E(Y))]$

$= E(XY) - E(X)E(Y)$

② $\rho_{xy} = \frac{Cov(X, Y)}{\sqrt{V(X)V(Y)}}$ $-1 \leq \rho \leq 1$

prove using

$V = X - E(X)$

$W = Y - E(Y)$

$0 < E(V + tW)^2$

Continuous.

① $f(x, y) \geq 0 \quad \forall\ x, y \in \mathbb{R}$

② $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)\, dx\, dy = 1$

③ $g(x) = \int_{-\infty}^{\infty} f(x, y)\, dx$ ($\sum$ fix)

Marginals

$h(y) = \int_{-\infty}^{\infty} f(x, y)\, dx$ ($\sum$ fix)

$g(x|y) = \frac{f(x, y)}{h(y)} \qquad h(y) > 0$

$h(y|x) = \frac{f(x, y)}{h(x)} \qquad h(x) > 0$

Binomial

① $P(x) = {}^nC_x\, p^x q^{n-x}$

② $\mu = np$

③ $V(x) = npq$.

Geometric

① $P(x) = pq^x$ $x = 0, 1, 2 \ldots$

② $\mu = 1/p$ $p \to$ success

③ $V(x) = q/p^2$ $q \to$ failure

$p + q = 1$.

Poisson

① $P(x) = \frac{m^x e^{-m}}{x!}$

↓ parameter.

$m = np \cong$ constant

$n \to \infty$

$p \to 0$

② $\mu = m$

③ $V(x) = m$.

Uniform

① $f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{elsewhere} \end{cases}$

② $\mu = \frac{a+b}{2}$

③ $V(x) = \frac{(a-b)^2}{12}$

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BASICS

(91) There are 21 tickets (1, 2, 3 . . . 21) drawn at random. What is the prob. that chosen ticket nos. are in AP?

(92) Btwn A & B, A speaks truth 75% of the time, B speaks truth 80% of the time. What is the prob. that A and B contradict each other?

(93) man alternatively tosses a coin and a die [beginning with a coin]. What is the prob. that he will get a head before tossing 5 or 6 on the die (till first head appears)

(94) A and B throw alternatively with a pair of dice:
i) A wins if he throws 6 before B throws 7.
ii) B wins if he throws 7 before A throws 6.

If A begins first, find A's chances of winning?

(95) A random experiment of tossing a pair of die is performed. A random variable X on this S.S is defined to be the sum of 2 nos. turning up. Find prob. dis, mean & variance.

(96) A continuous r.v has pdf given by $f(x) = kx^2$, $0 \le x \le 1$. Find k

Find $P(X < 1/2)$
$P(X \ge 3/4)$

(97) $P(A) = 3/4$
$P(B) = 3/8$
show that:
(i) $3/4 \le P(A \cup B)$
(ii) $1/8 \le P(A \cap B) \le 3/8$
(iii) $3/8 \le P(A \cap \bar{B}) \le 5/8$

(98) Digits 1, 2, 3, . . . n are written in order. What is the probability that no digit is occupying its proper place.

BERNOULLI TRIALS.

(99) Two dice are thrown 10 times in succession
What is the probability of obtaining a double six at least once?

(910) Find prob of correctly guessing at least 6 out of 10 answers in a
(1) T/F exam
(2) multiple choice with 4 options, 1 correct.

(911) A pays 1 $ for each participation in the following game.
3 dice are thrown ⇒ if 1 Ace ⇒ 1 $, 2 Ace ⇒ 2 $, 3 Ace ⇒ 8 $ } else nothing

(1) Is the game fair??
(2) If not, how much should player receive when 3 Ace appear to make the game fair?

CDF

(912) The pdf of a r.v. X is $f(x) = \begin{cases} x, & 0 \le x \le 1 \\ 2-x, & 1 \le x \le 2 \\ 0, & \text{otherwise} \end{cases}$
compute cdf of X.

(913) If the pdf of a r.v. X is $f(x) = \begin{cases} 20x^3(1-x), & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$
find cdf of f.

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TWO-DIM RANDOM VARIABLES.

Q14) Two product lines manufacture a certain type of item. Suppose that the capacity is 5 items for Line I & 3 items for line II. Assume that no. of items actually produced by the machines is a random variable.

Let $B = \{(X,Y) \mid$ X produced more than Y$\}$

Find $P(B) = ??$

Y \ X	0	1	2	3	4	5
0	0	0.01	0.03	0.05	0.07	0.09
1	0.01	0.02	0.04	0.05	0.06	0.08
2	0.02	0.03	0.05	0.05	0.05	0.06
3	0.01	0.08	0.04	0.06	0.06	0.05

Q15). 2 dim r.v has joint pdf given by

$$f(x,y) = \begin{cases} x^2 + xy/3 & 0 \le x \le 1,\ 0 \le y \le 2 \\ 0 & \text{elsewhere} \end{cases}$$

Find $P(B)$, $B = \{X + Y \ge 1\}$

Q16) Suppose 2 dim r.v has joint pdf given by

$$f(x,y) = \begin{cases} kx(x-y) & 0 < x < 2,\ -x < y < x \\ 0 & \text{elsewhere} \end{cases}$$

① find k
② marginals w.r.t x & y.

Q17) 2 dim r.v has joint density funcn

$$f(x,y) = \begin{cases} 6(e^{-2x-3y}) & , x, y \ge 0 \\ 0 & , \text{elsewhere} \end{cases}$$

Find: ① $P(1 < x < 2, 2 < Y < 3)$
② $P(0 < x < 2, Y > 2)$
③ marginals
④ conditional densities.

Q18) The coefficients a,b,c of a quadratic equation $ax^2 + bx + c = 0$ are selected at random by throwing a die 3 times. What is the prob. of the roots of the eqn being real?

DISTRIBUTIONS, MARGINALS, CORR. COEFF

Q19) Coin tossed 4 times Find P(exactly one head)
P(at most 3 heads)
P(at least 2 heads)

Q20) A & B each hit a shot independently until each had hit its own target P(hitting target) at each shot are (3/5) & (5/7) respectively. Find prob. that B will require more shots than A.

Q21) A die is thrown independently until 1 appears. Assume P(getting 1) = p. Find value of p such that prob. of odd no of throws required is equal to 0.6.

Q22) A switch board can handle only 4 tele calls/min. If incoming call/min follows a poisson distribution with parameter 3. Find the probability that switch board is over toned in any one minute.

Q23). In quality control, m = 0.2 rejects are done per minute on the average. Find prob. of rejecting: ① one in 3 mins ② at least 2 in 5 mins ③ at most 1 in 15 mins

If r.v.k is uniformly distributed over (0,5)
Prob(roots of eqn: $4x^2+4xk+(k+2)$) are real?

Q25) On a certain city transport route buses ply every 30 mins b/w 6 to 10 am. If a person reaches the bus stop on this route at a random time during this period, what is the P(he will have to wait for at least 20 minutes)=?

Q26) Passenger arrives at a bus stop at 10 AM, knowing that bus arrives at some time b/w 10 & 10:30 AM. What is the prob. that he will have to wait longer than 10mins. If at 10:15 AM, the bus didn't arrive yet, what is the probability that he will have to wait atleast 10 more minutes

Q27) Let X be the outcome when a fair die is tossed, ① find mgf of x.
② using mgf, find mean and variance.

Q28) Let X be a r.v & $P(x)=ab^x$ where a,b are +ve, $a+b=1$, $x=0,1,2\ldots$
Find mgf of X. Show that $m_2=m_1(2m_1+1)$ $m_1 \to$ 1st moment
$m_2 \to$ 2nd moment.

Q29) Suppose we select one point at random, within circle of radius R with centre at origin and (X,Y) be coordinates of the point chosen from $x^2+y^2 \le R^2$, then (X,Y) is a 2 dim r.v with joint pdf $f(x,y)=\begin{cases} k & x^2+y^2 \le R^2 \\ 0 & x^2+y^2 > R^2 \end{cases}$
① Find k
② Marginal pdf of x & y
③ P(dist from origin to point $\le a$)

Q30) Suppose that the 2 dimensional random variable (X,Y) is uniformly dis. over the Δular region $R=\{(x,y) \mid 0<x<y<1\}$ Find ρ (corr coeff)

Q31) $f(x,y)=\begin{cases} x+y, & 0<x<1,\ 0<y<1 \\ 0, & \text{elsewhere} \end{cases}$ Compute ρ.

Q32) mgf of a r.v. $X = M_x(t)=(0.4e^t+0.6)^8$. What is the mgf of r.v $Y=3x+2$

Q33) r.v. X pdf has $f(x)=\frac{1}{2}e^{-|x|}$ $-\infty<x<\infty$. Obtain mgf of x and find E(x), V(x).

Q34) $f(x)=\begin{cases} x/2 & 0\le x\le 2 \\ 0 & \text{elsewhere} \end{cases}$ find mgf. (Q35) 2 r.v $X_1=\frac{1}{2}$, $X_2=\frac{-1}{2}$, pdf $=\frac{1}{2}$ Find mgf.

Q36) Two independent r.v X_1 & X_2 had mean 5, 10 & variance 4, 9
Find corr(U,V) where $U=3x_1+4x_2$
$V=3x_1-x_2$.

Q37) Find the c.d.f for exponential distribution.

Q38) In a certain town, duration of shower is exponentially distributed with mean=5mins
What is the prob. that: shower will last for ① 10 mins or more
② less than 10 minutes.

Q39) Inc in sales $\Rightarrow$ exponentially distributed with mean=800
If sales tax is levied at rate of 6%. Find the probability that increase in sales tax returns exceed 30/- day.

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Q40) Life of a bulb is advertised to have a mean of 100 hours
Assume exponential dis. Find $P(x<200 \text{ hrs})$ $P(100<x<300 \text{ hrs})$ x: life of bulb.

Q41) Avg daily turnout in dept store ₹10000, net profit 8%. Assume turnout has expo. dis.
find P(net profit exceeds 3000 on 2 consecutive days)=?

Q42) mean = 1000 hrs, exp. distribⁿ } P(bulb lasts 2000 hrs | even when it has already lasted 1000 hrs.

NORMAL DISTRIBUTION.

Q43) If $X \sim N(2, 0.16)$ find ① $P(X \geq 2.3)$ ② $P(1.8 \leq X \leq 2.1)$

Q44) X: temp variable, normally distributed, expectation 50°, variance 4° } $P(48<X<53)$=?

Q45) diameter of electric cable is normally distributed with mean 0.8 and variance 0.0004. P(diameter exceeds 0.81 inch)?

Q46) s.d of certain grp of 1000 high school students' grades was 11%.
mean grade 78%. Assume normal distribution
① how many grades above 90%, ② below 60%, ③ btwn 75% and 85%.

Q47) Steel rods are manufactured to be 3" in diameter but are acceptable if they are inside 2.99" & 3.01". It is observed that 5% are rejected due to oversize, 5% are rejected due to undersize.
Assume diameters to be normally distributed.
① Find std deviation
② Evaluate proportion of rejects if permissible limits are extended to 2.985" and 3.015"

Q48) Normal Distribution; $X<45 \Rightarrow 31\%$, $X>64 \Rightarrow 8\%$ } find μ, σ.

Q49) Show that $E((X-\mu)^{2n}) = 1.3.5.7 \ldots (2n-1)\sigma^{2n}$ for normal distribution with mean $=\mu$, s.d $=\sigma$, var $=\sigma^2$

INEQUALITIES

Q50) X is a r.v = gain in Rs. with $E(X)=$₹10. Find the upperbound for the chance that person wins (gains) at least ₹50.

Q51) avg. delivery time ($\mu=2$hrs), $\sigma=0.5$ hrs. What is the maximum probability that a delivery will take 3hrs or more?

Q52) no. of items produced in a factory during a week is a random variable with $\mu=50$ ① What can be said abt P(week's producⁿ will exceed 75)
② If var = 25, then prob. producⁿ will be between 40 & 60.

Q53) A symmetric die is thrown 600 times. Find the lower bound for prob. of geting 80 to 120 sixes.

Suppose that the no. of motor cars arriving in a certain parking lot in any 15 mins period obeys a poisson distribution with mean 80. Use chebyshev's inequality to determine a lower bound for the prob that no. of motor cars arriving in a given 15 mins period will be btwn 60 to 100.

Q55) The duration of spell of rain in a certain place is a gamma variate with parameter $\alpha=2$, $\beta=10$. Find the probability that on a particular day, the duration of spell of rain is less than 30 minutes.

Q56) The daily sales of a newspaper in a city is a gamma variate with mean 100 and variance 5000. Find the prob. that sales exceed 150 nos. on a particular day?

Q57) Let X be a c.r.v with pdf $f(x)=e^{-x}$, $x>0$

$Y=2X+1$
$R_X=\{x>0$ } compute $P(Y\geq 5)$.

Q58) Let $X=1,2,3\ldots$
Suppose $P(X=n)=\left(\frac{1}{2}\right)^n$

$Y=\begin{cases}1 & \text{if } X=\text{even}\\ -1 & \text{if } X=\text{odd}\end{cases}$

Find $P(Y)$

Q59) X has pdf $f(x)=\begin{cases}2x & 0<x<1\\ 0 & \text{elsewhere}\end{cases}$

$Y=H(X)=3X+1$, pdf of $Y=?$.

Q60). $f(x)=\begin{cases}2x & 0<x<1\\ 0 & \text{elsewhere}\end{cases}$ $Y=e^{-x}$ } find pdf.

Q61) Let X be r.v with pdf $f(x)$ defined for $-\infty<x<\infty$. Show that pdf of $Y=x^2$ is $\frac{1}{\sqrt{2y}}\left(f(\sqrt{y})+f(-\sqrt{y})\right)$

Q62) Let X be a r.v with pdf $f(x)=\begin{cases}1/2 & -1\leq x\leq 1\\ 0 & \text{elsewhere}\end{cases}$ find pdf of $y=x^2$.

Q63) X: r.v with pdf $f(x)=\begin{cases}2x & 0\leq x\leq 1\\ 0 & \text{elsewhere}\end{cases}$ Find pdf of $Y=8X^3$.

Q64) Let X: uniform distribution over $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$. Obtain pdf of $Y=\tan X$.

Q65) X: r.v pdf $f(x)=\begin{cases}x^2/9 & 0\leq x\leq 3\\ 0 & \text{elsewhere}\end{cases}$ find pdf of $Y=X^3$.

Q66) Suppose $X\sim U(-1,1)$ Let $Y=4-X^2$ find pdf of Y.

FUNCTIONS OF 1D, 2D RANDOM VARIABLES.

Q67) If X and Y are independent random variable each foll. $N(0,2)$ then find pdf of $Z=2X+3Y$.

Q68) If X and Y follow the exponential distribution with parameter 1 and are independent. Find pdf of $U=X-Y$.

Q69) If the joint pdf of (X,Y) is given by $f_{xy}(x,y)=x+y$ $0\leq(x,y)\leq 1$
Find pdf of $U=XY$

Q70) Prove that $Z=X/Y$ follows a Cauchy Distribution. $X\sim N(0,2)$ $Y\sim N(0,2)$

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